

## Lecture 2: Form of the Summation Operator

$$\begin{aligned} S &= \nabla^{-1} \\ &= (1-B)^{-1} \\ &= \frac{1}{1-B} \end{aligned}$$

$$\begin{aligned} \text{Let } f(B) &= (1-B)^{-1} \\ \Rightarrow f'(B) &= (1-B)^{-2} \\ \Rightarrow f''(B) &= 2(1-B)^{-3} \\ \Rightarrow f'''(B) &= 3 \times 2 (1-B)^{-4} \end{aligned}$$

Generally, if  $f^{(k)}(B)$  is the  $k^{\text{th}}$  derivative of  $f(B)$  then

$$f^{(k)}(B) = k! (1-B)^{-(k+1)}$$

The Taylor series expansion of  $f(B)$  about 0 is

$$\begin{aligned} f(B) &= f(0) + f'(0)B + \frac{f''(0)}{2!} B^2 + \dots \\ &= 1 + B + B^2 + \dots \end{aligned}$$

$$\therefore S = \nabla^{-1} = \sum_{j=0}^{\infty} B^j$$